Understanding the Last State in TD Learning – Does It Learn?

The **last state (final destination)** in TD Learning **does not learn in the same way as other states** because:

1. **It has no next state** → There is no future reward to consider.
2. **Its utility is equal to its reward** → No updates are needed.

Why Doesn't the Last State Learn?

- **Other states learn** by looking at **both immediate and future rewards**.
- The **last state only gets an immediate reward**, so its utility is **always fixed**.

Formula for last state U(F)U(F):

U(F)=r(F)U(F) = r(F)

Since there is no **next state** s′s', the TD formula simplifies to just the immediate reward.

Example: Food Court (F) in Our Problem

📌 **Food Court (F) is the last stop**. The reward is **10**, so:

U(F)=10.0U(F) = 10.0

Even after many iterations, **this value never changes** because there's no future reward to learn from.

Does the Last State Ever Change?

- **If the reward for the last state changes**, then its value will change.
- Otherwise, it stays fixed throughout learning.

Key Takeaway in Simple Words

✓ The last state doesn't "learn" because it has no future state to learn from.
✓ Its value is always equal to its reward.
✓ Other states update their values by looking at future rewards, but the last state has no future to consider.